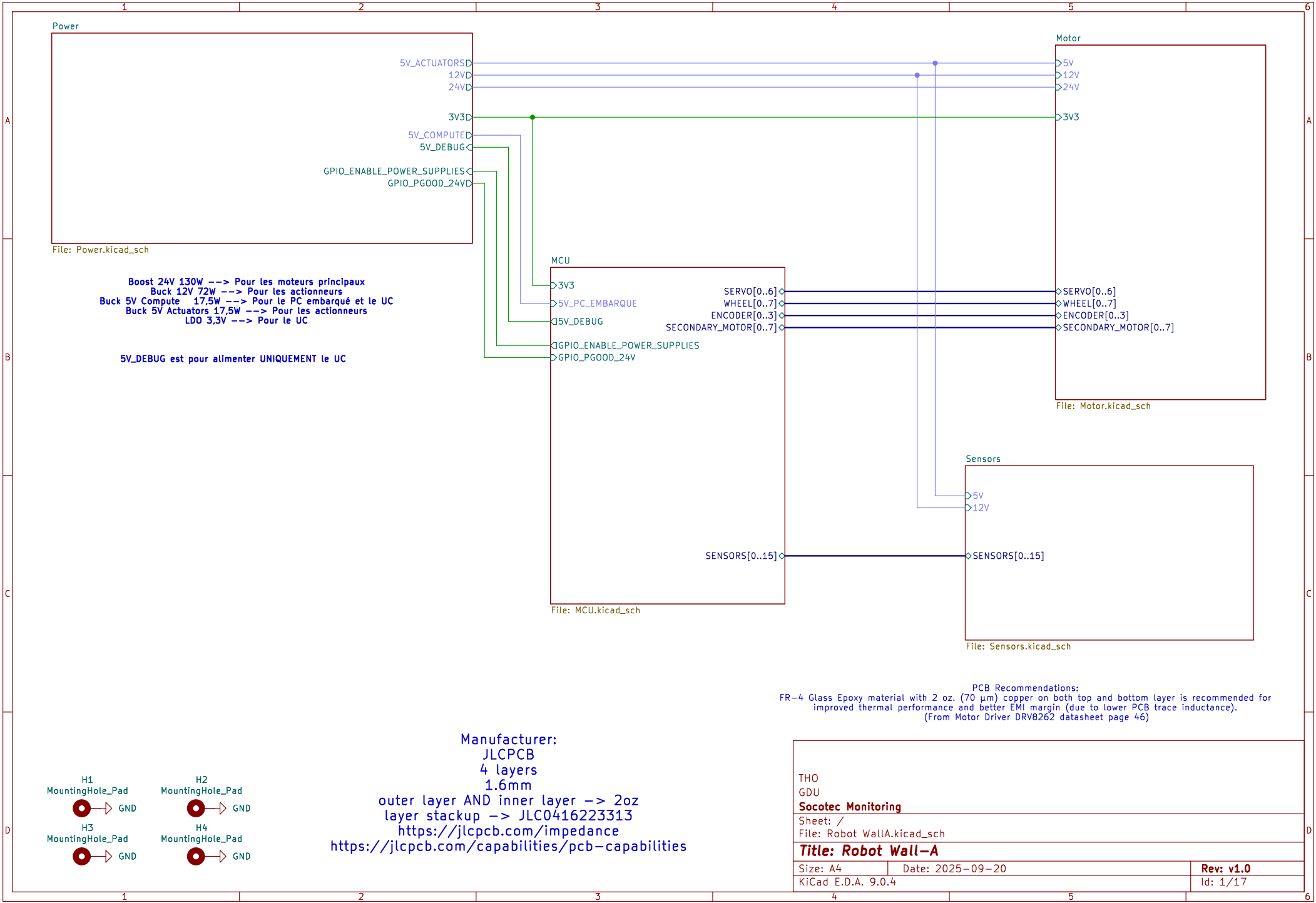
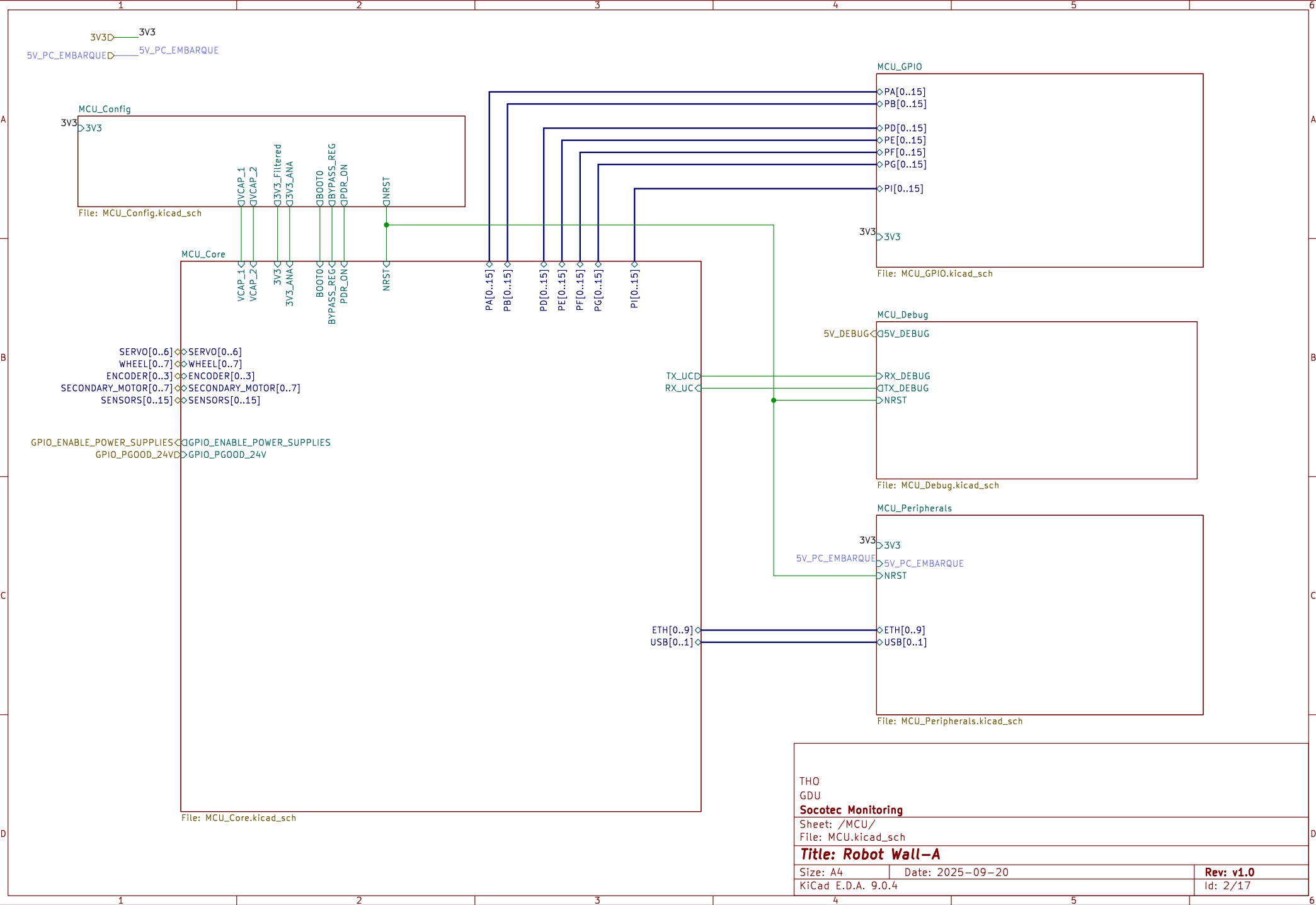






TH0
GDU

Socotec Monitoring

Sheet: /MCU/
File: MCU.kicad_sch

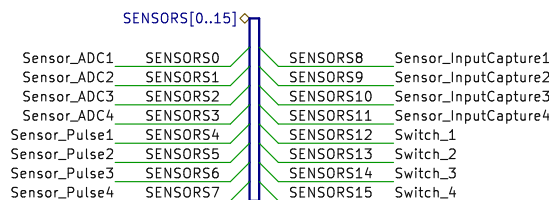
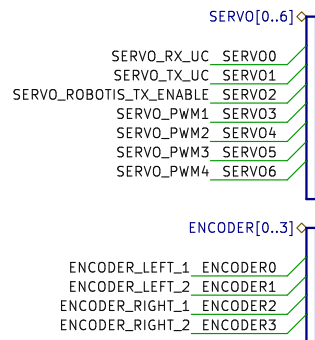
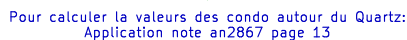
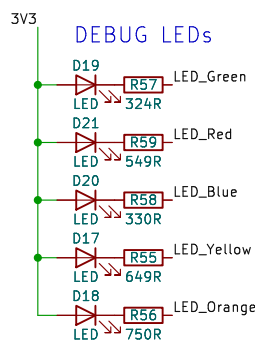
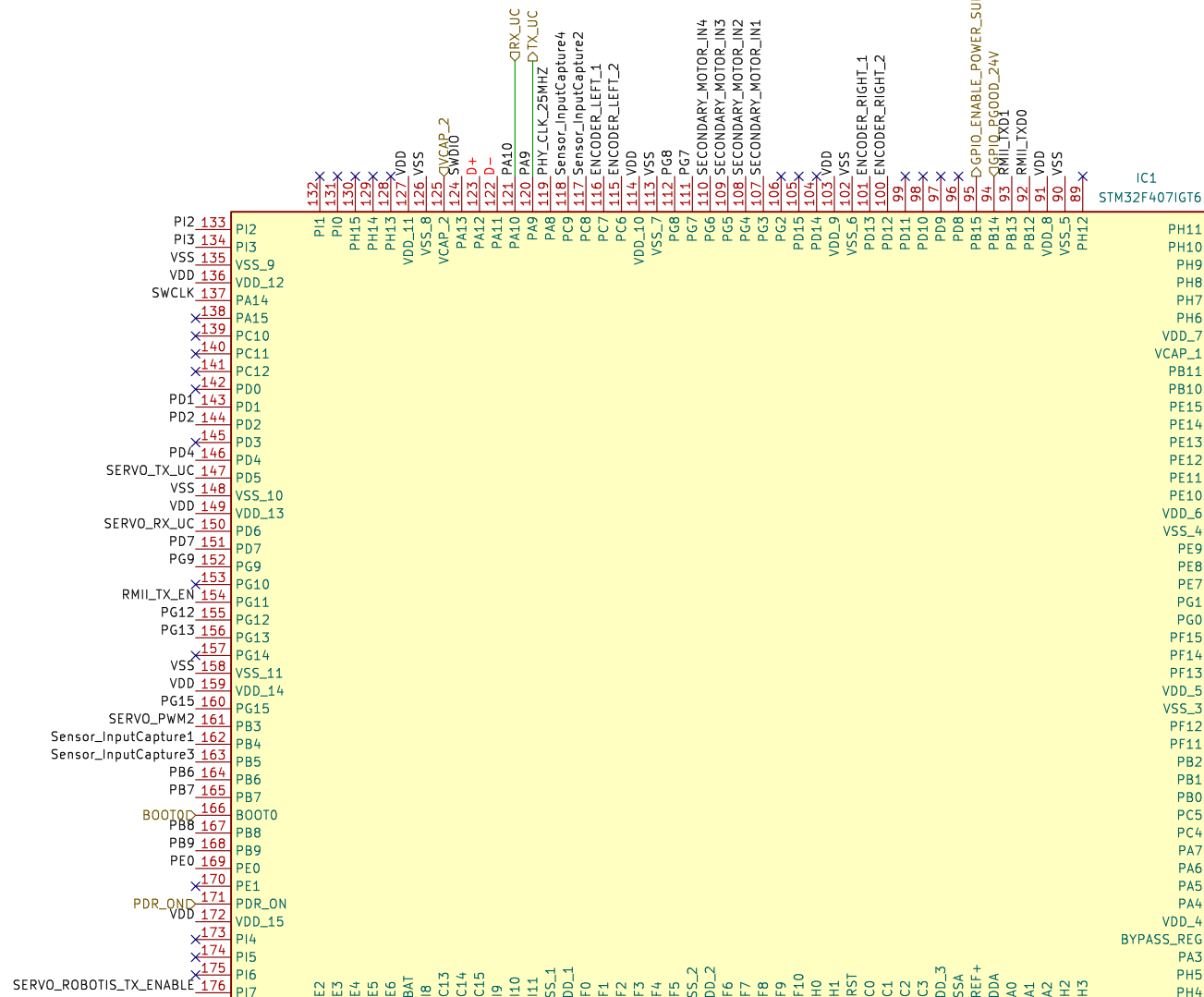
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

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Id: 2/17



Title: Robot Wall-A

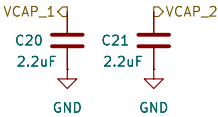
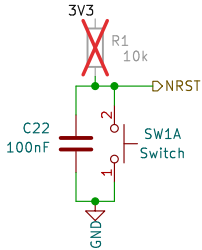
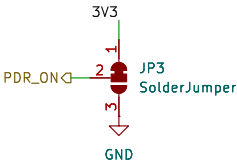
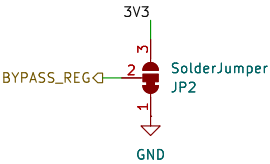
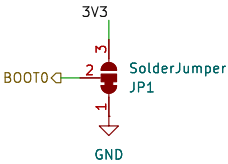
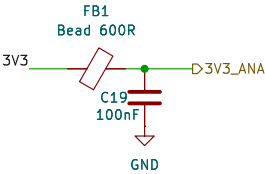
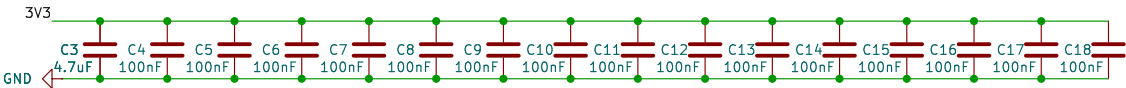
Rev: v1.0

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8

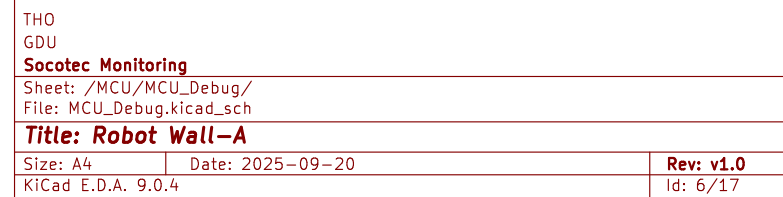
Must be as close as possible from μC

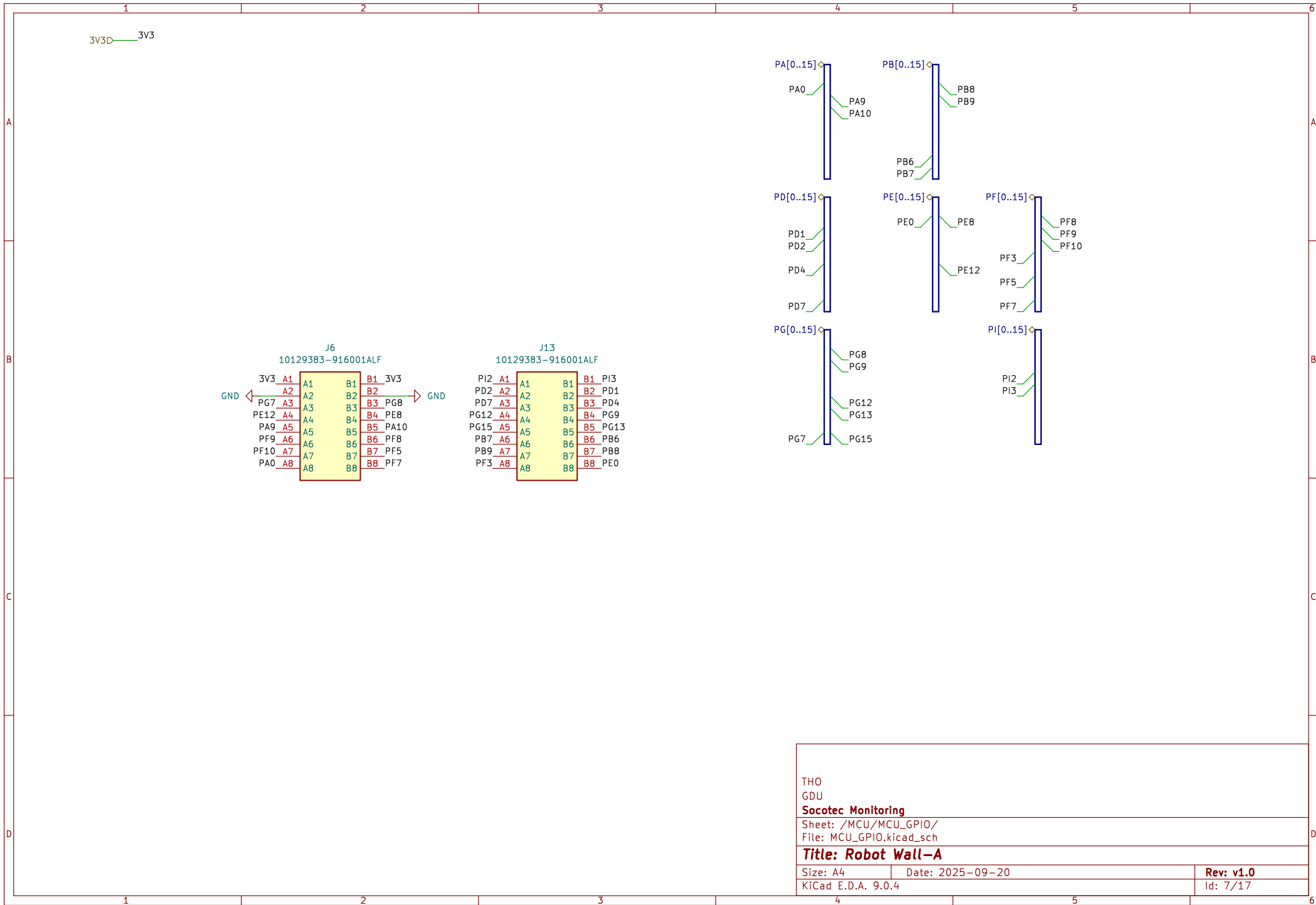
3V3 $\rightarrow$ 3V3_Filtered

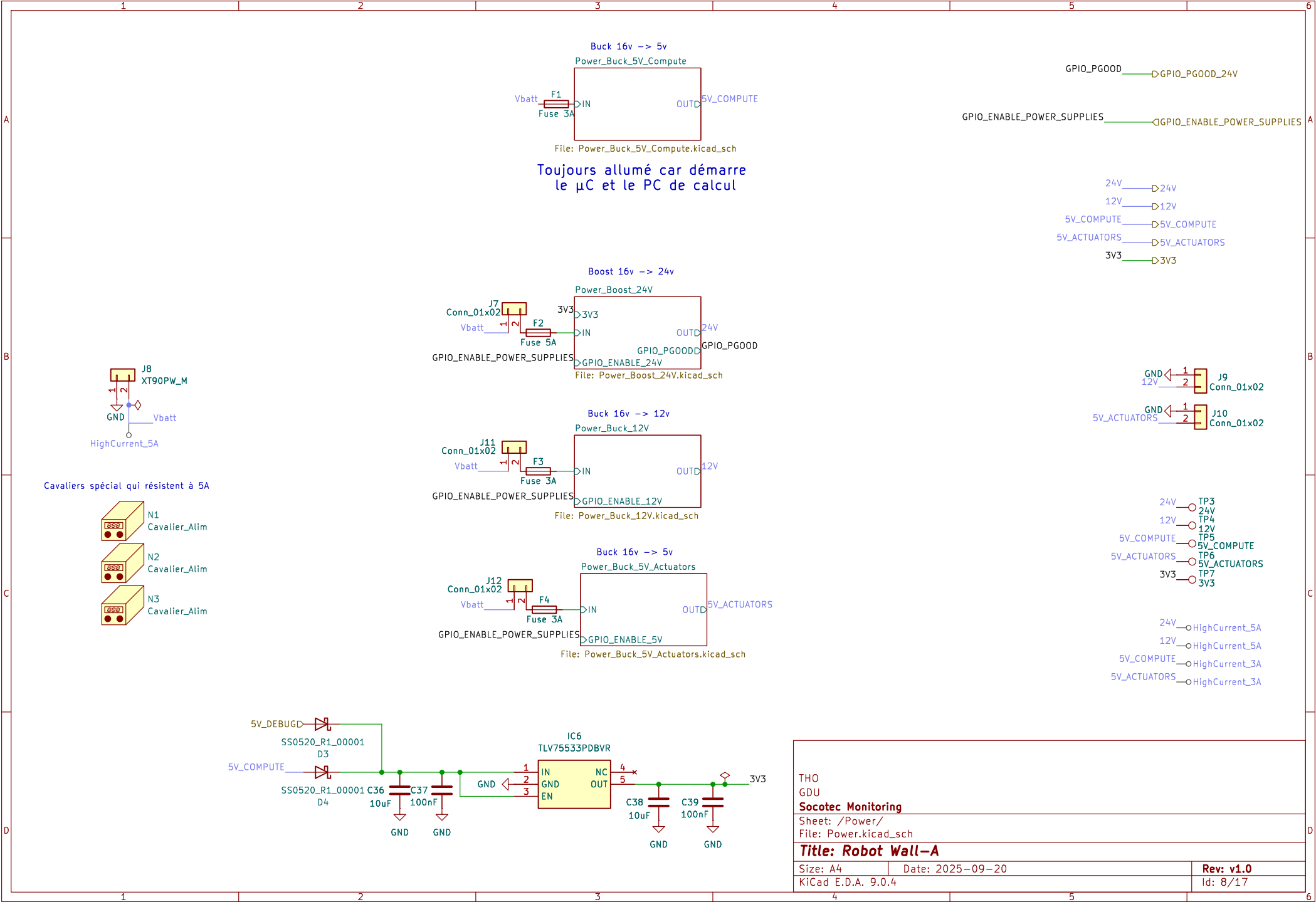


TH0		
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Socotec Monitoring		
Sheet: /MCU/MCU_Config/		
File: MCU_Config.kicad_sch		
Title: Robot Wall-A		
Size: A4	Date: 2025-09-20	Rev: v1.0
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[illegible][illegible]

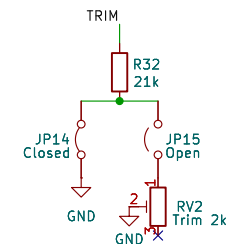
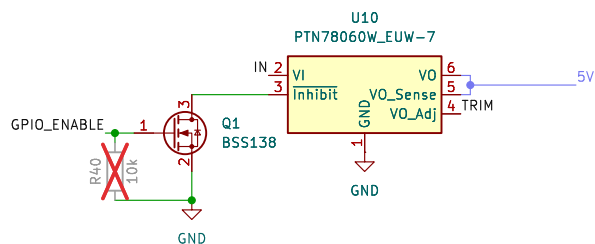
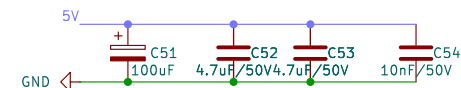
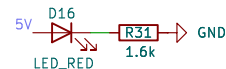
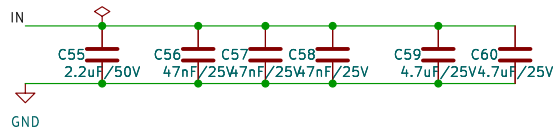






IND IN

5V OUT



TH0
GDU

Socotec Monitoring

Sheet: /Power/Power_Buck_5V_Compute/
File: Power_Buck_5V_Compute.kicad_sch

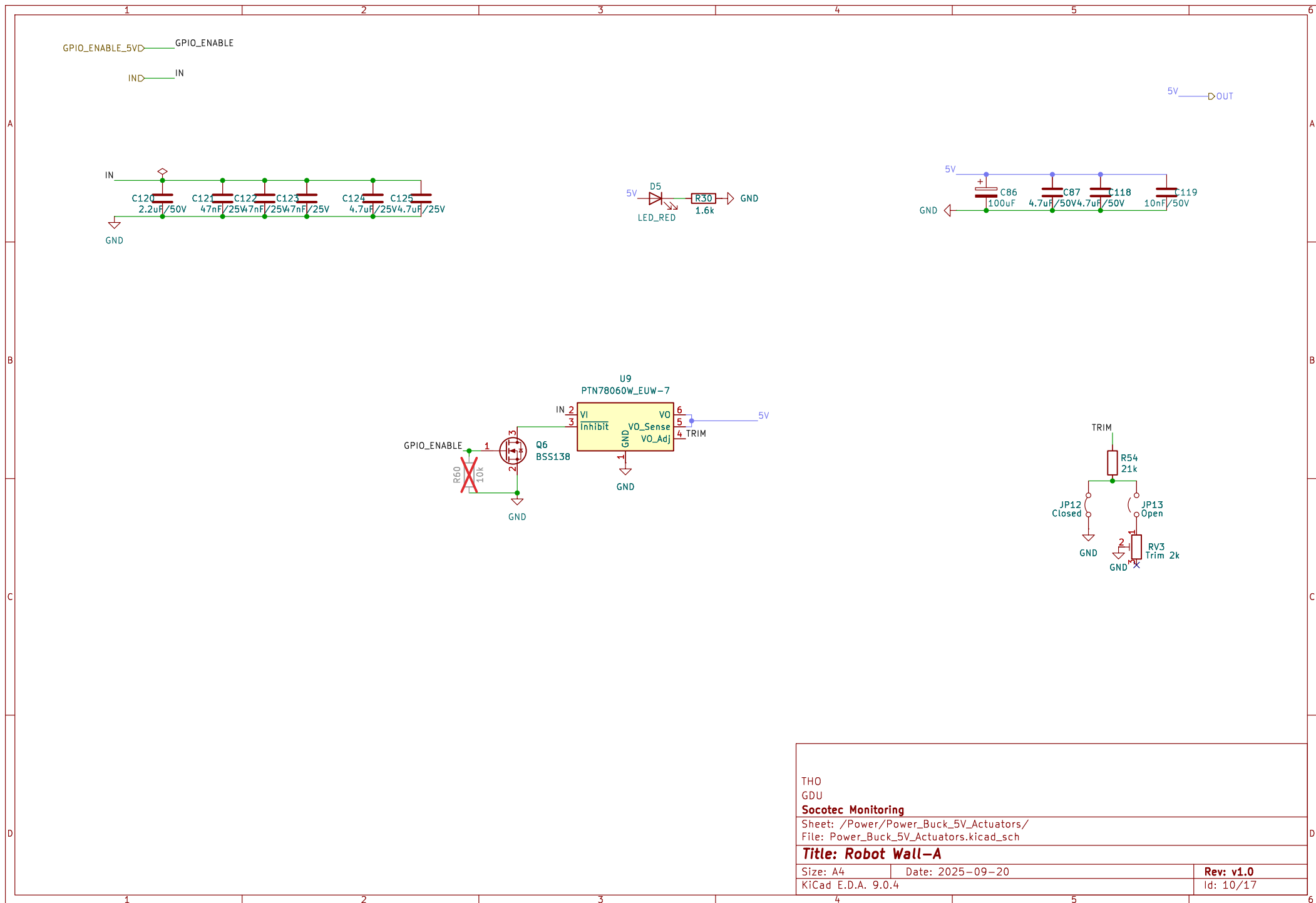
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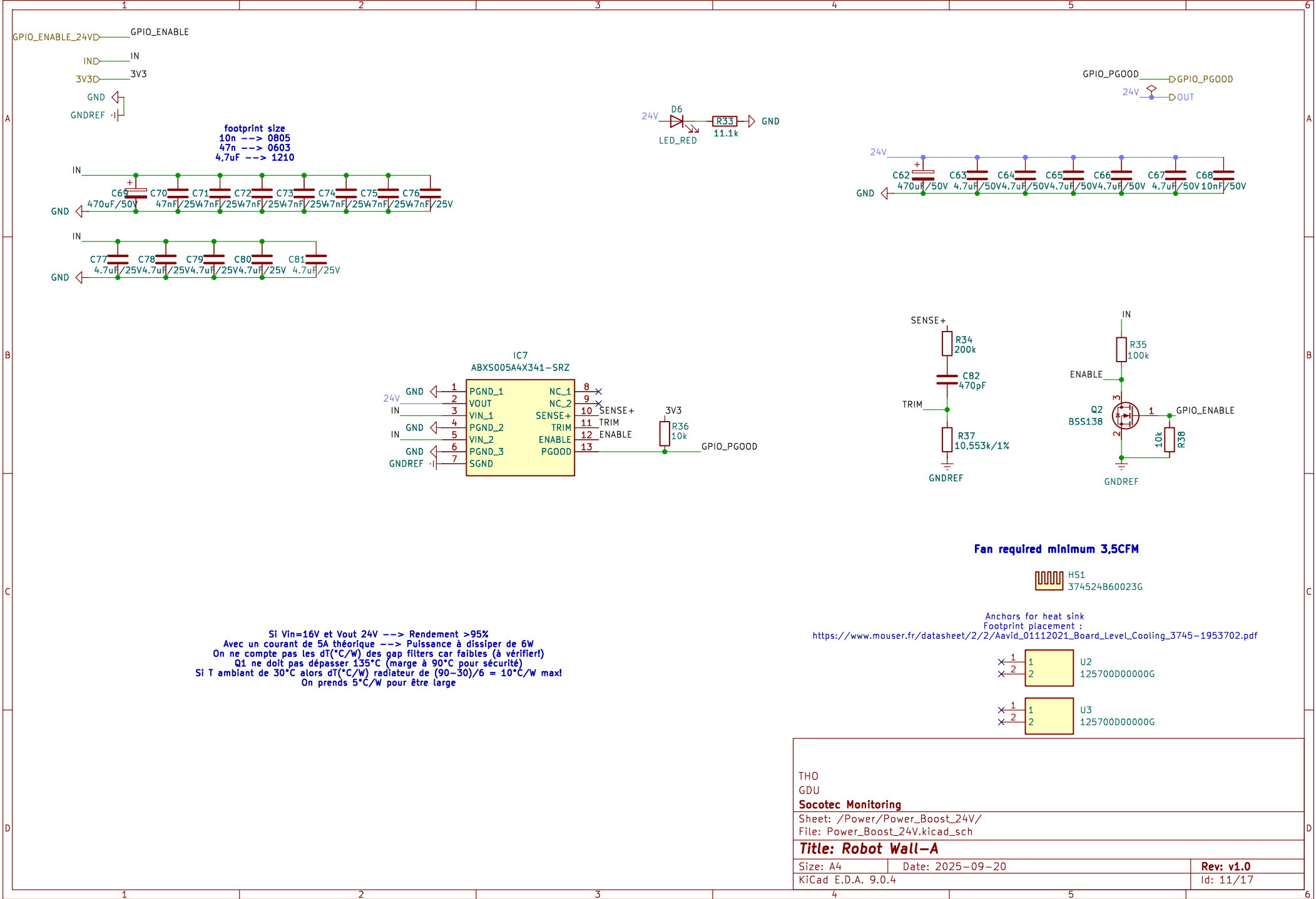
Size: A4 Date: 2025-09-20

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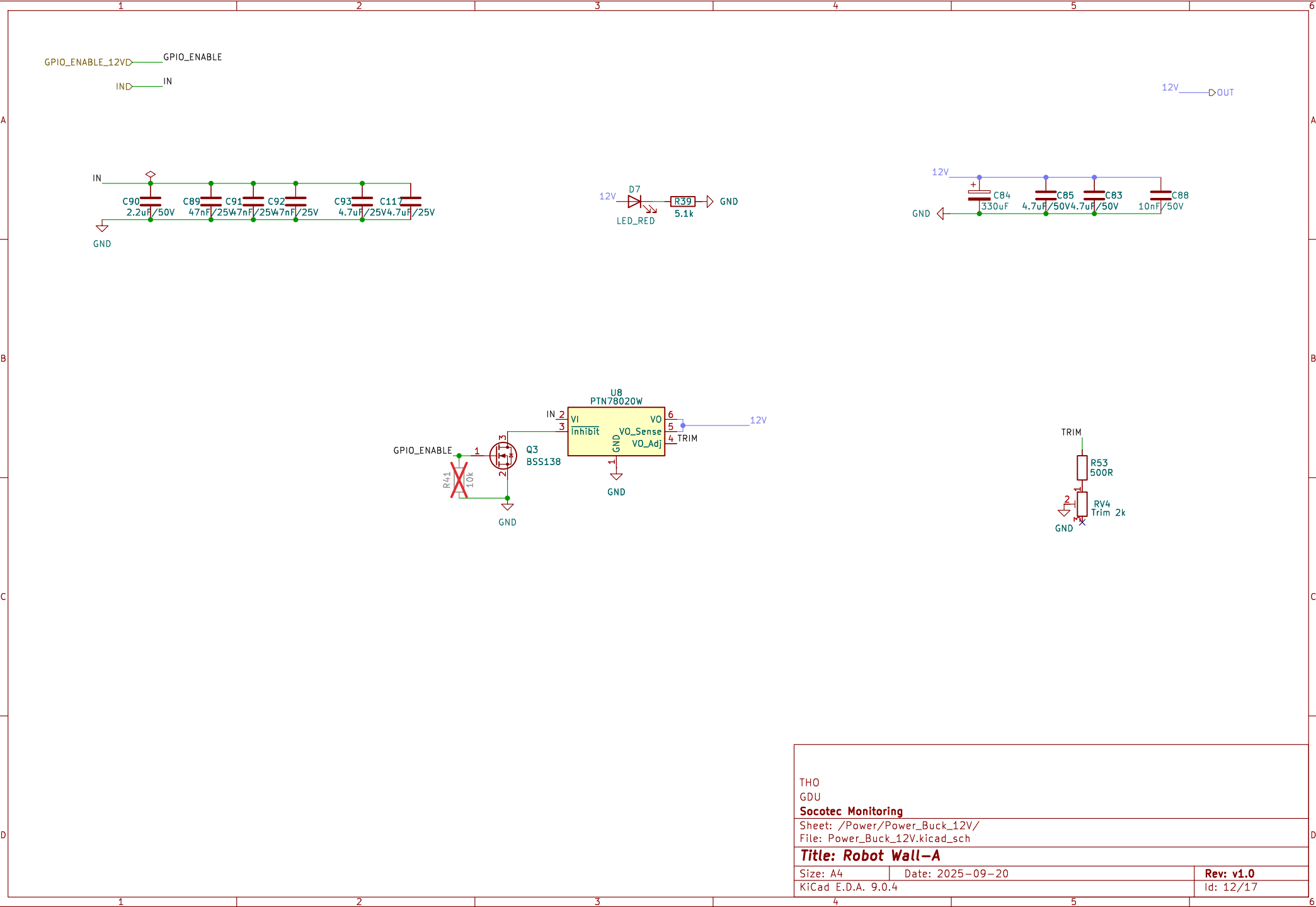
Rev: v1.0

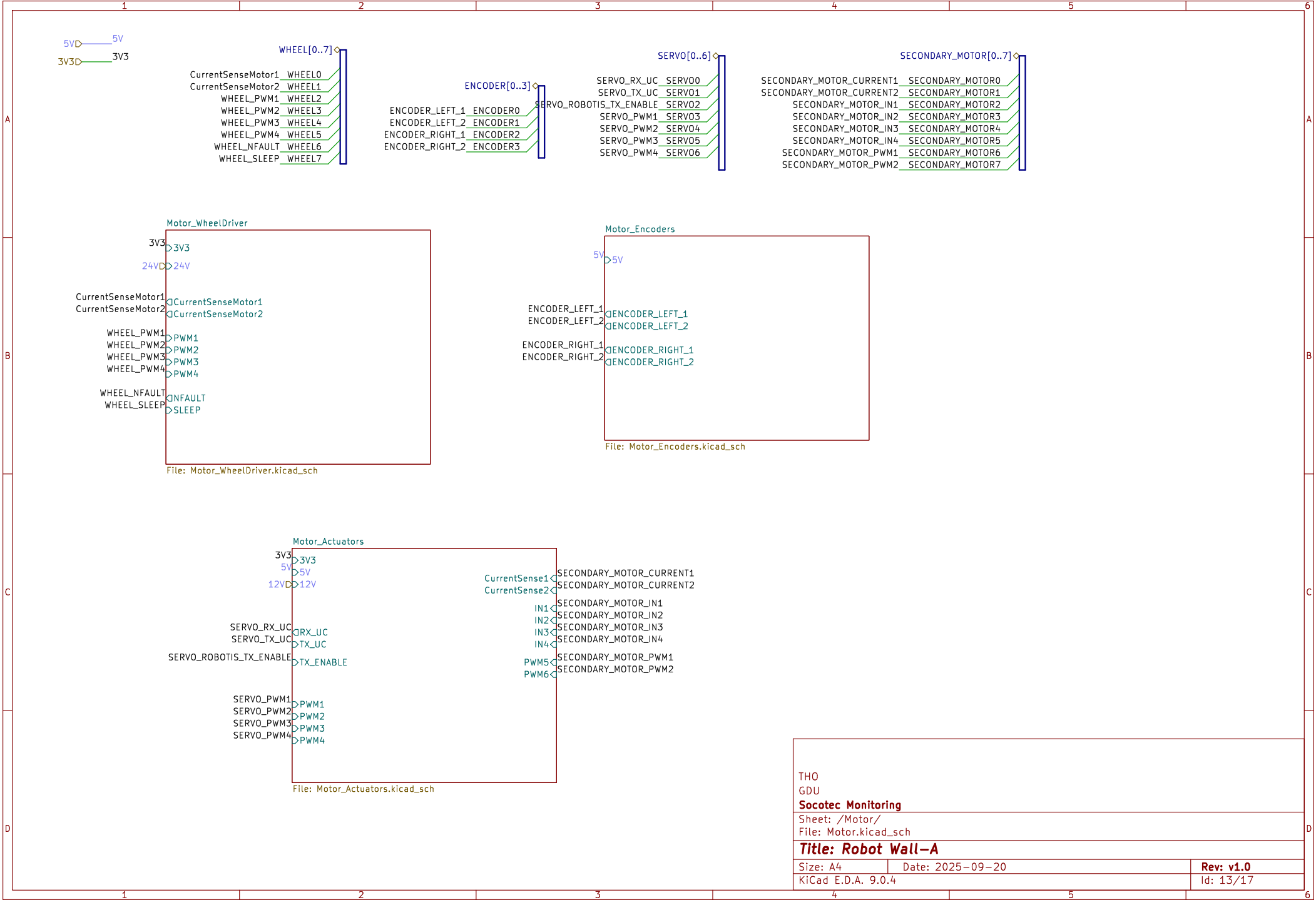
Id: 9/17





Si Vin=16V et Vout 24V --> Rendement >95%
Avec un courant de 5A théorique --> Puissance à dissiper de 6W
On ne compte pas les dT(*C/W) des gap filters car faibles (à vérifier!)
Q1 ne doit pas dépasser 135°C (marge à 90°C pour sécurité)
Si T ambiant de 30°C alors dT(*C/W) radiateur de (90-30)/6 = 10°C/W max!
On prends 5°C/W pour être large



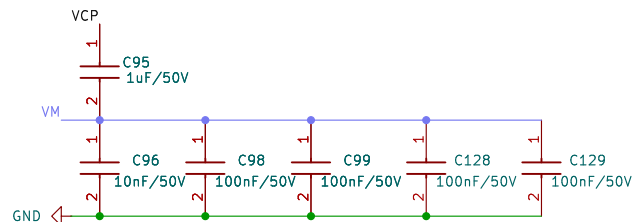
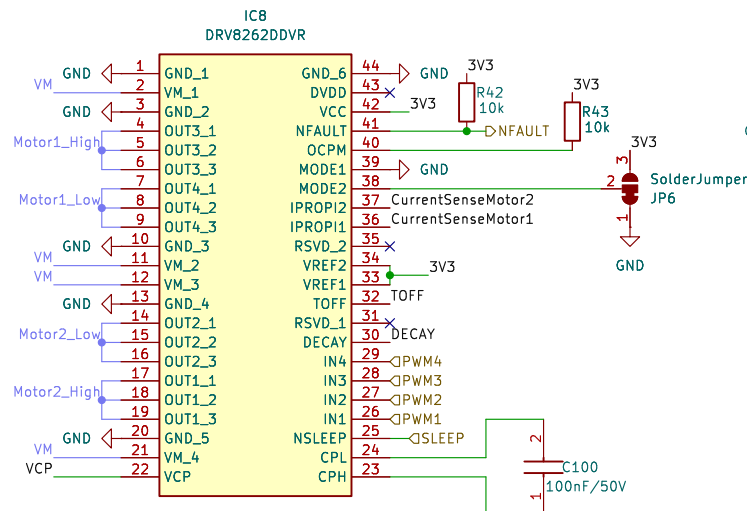


3V3D 3V3
24VD VM



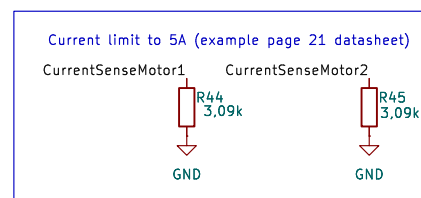
Default Configuration:
OCPM = 1 --> Over Current Protection error = Retry
MODE = 0 --> PH/EN mode
Toff = 1 --> Waiting time between change = 16µs
Decay = HZ --> Auto detect mode

CurrentSenseMotor1 --> CurrentSenseMotor1
CurrentSenseMotor2 --> CurrentSenseMotor2



Motor1_High 1 1 J16
Motor1_Low 2 2 39501-1002

Motor2_Low 1 1 J15
Motor2_High 2 2 39501-1002



ATTENTION!!
Pour simplifier le routage,
les phases du moteurs on été inversées.
Attention au câblage!

https://www.mouser.fr/datasheet/2/2/Aavid_01112021_Board_LeveL_Cooling_3745-1953702.pdf

U4
125700D00000G

U5
125700D00000G

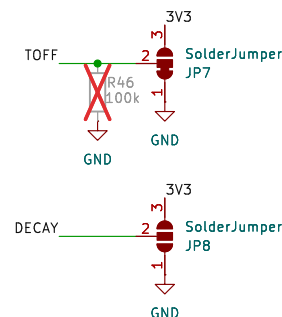
HS2
374524B60023G

Fan required minimum 3,5CFM

2x Transmotec PD4266-24-17-BFEC
24V / 2,1A nominal
4,2A nominal total

Power loss calculation and dissipator estimation
(page 33 to 35 datasheet)
Calculations already done in datasheet (same example with 4A)
Conclusion: Dissipator with less than 4°C/W

Heat sink possibilities:
WAVE-35-12 with anchors 125700D00000G --> Fan required minimum 3CFM
374524B60023G with anchors 125700D00000G --> Fan required minimum 3,5CFM
567-45AB with screws --> Fan required minimum 4CFM



Motor1_High --> HighCurrent_5A
Motor1_Low --> HighCurrent_5A
Motor2_High --> HighCurrent_5A
Motor2_Low --> HighCurrent_5A

TH0
GDU

Socotec Monitoring

Sheet: /Motor/Motor_WheelDriver/
File: Motor_WheelDriver.kicad_sch

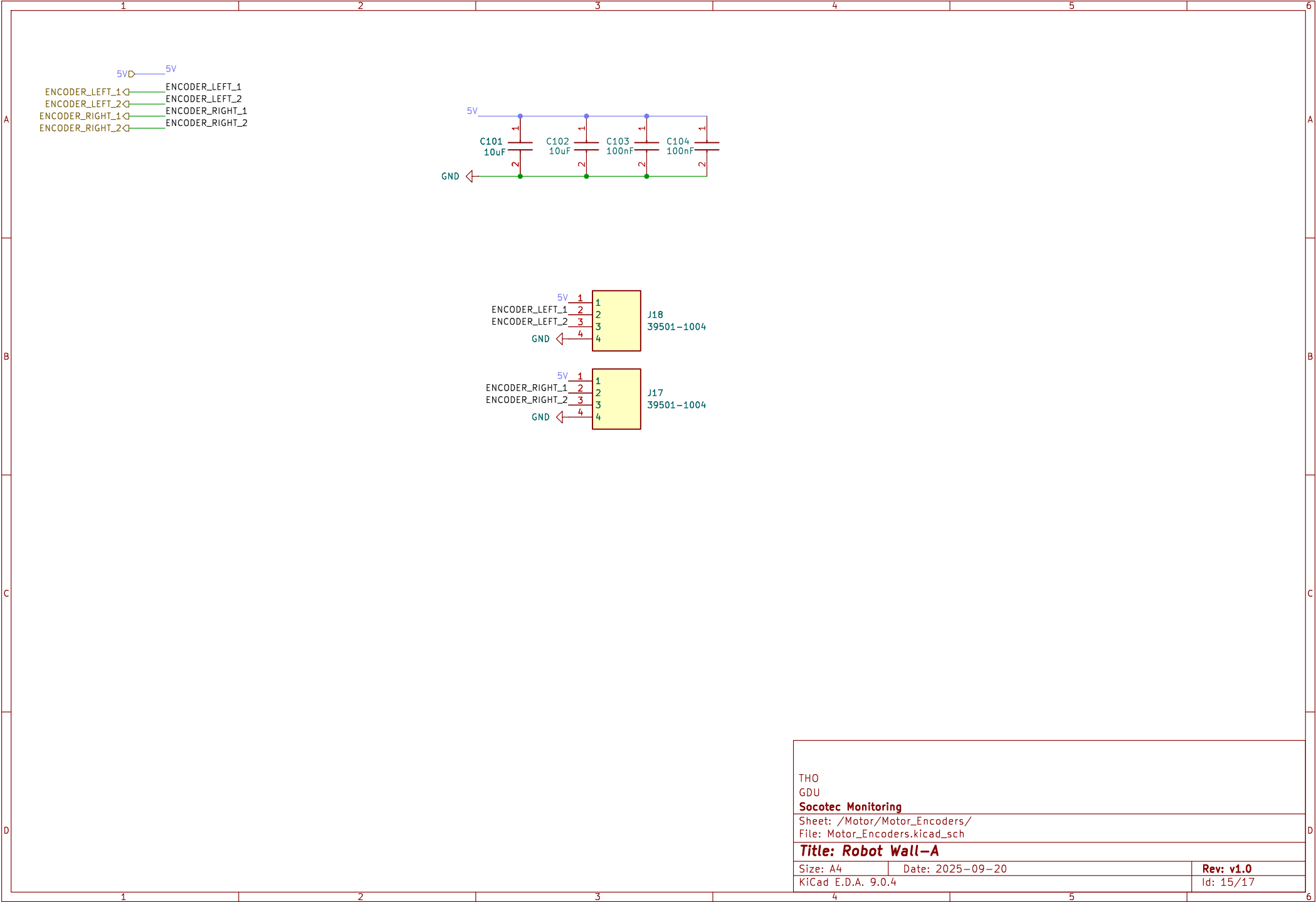
Title: Robot Wall-A

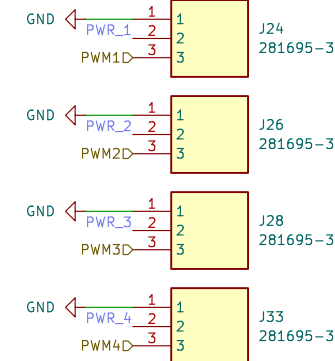
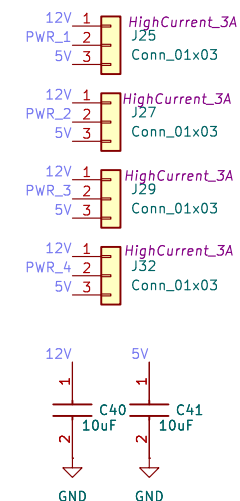
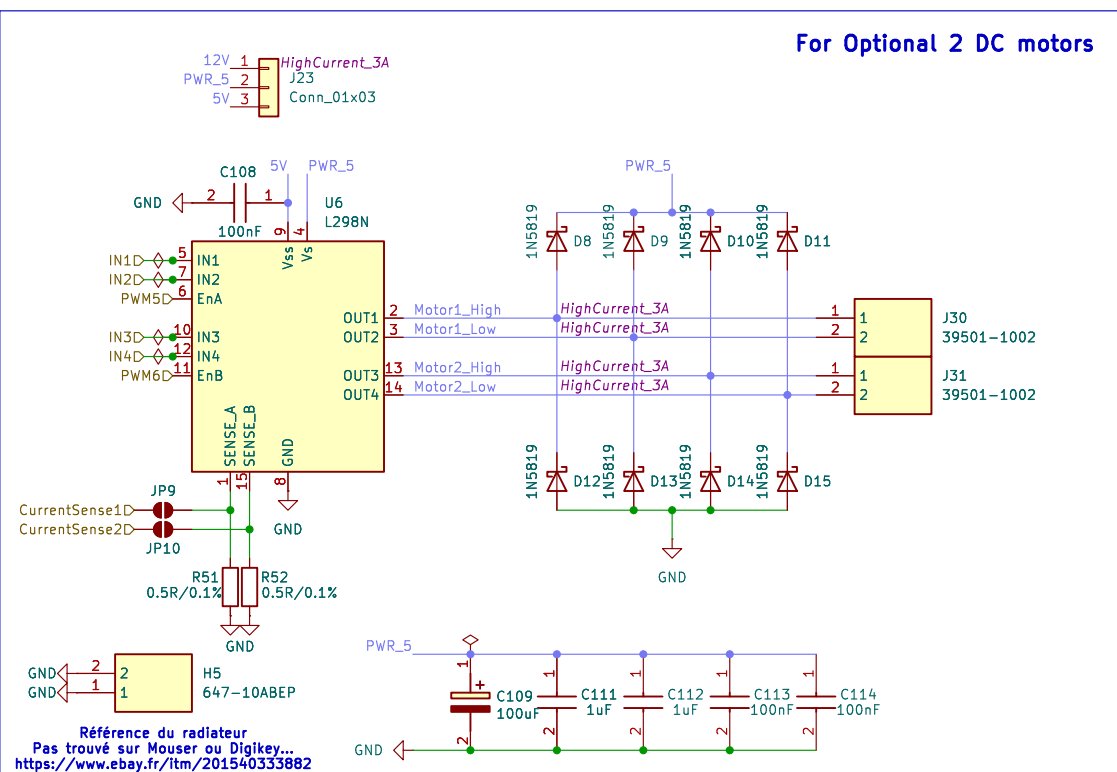
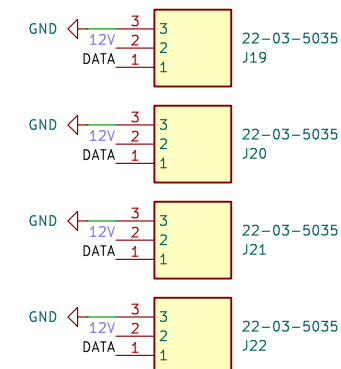
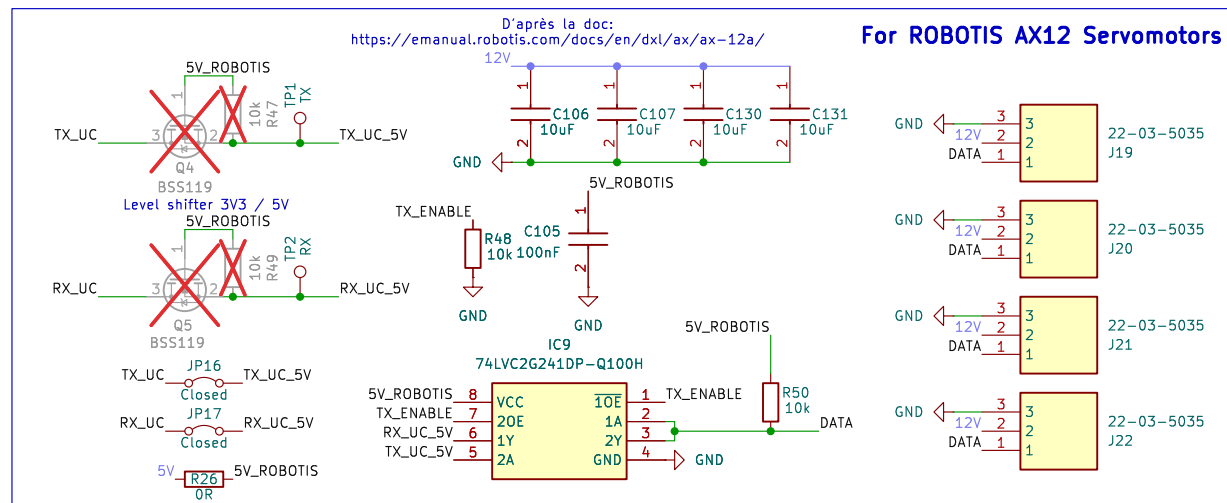
Size: A4 Date: 2025-09-20

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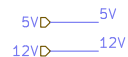
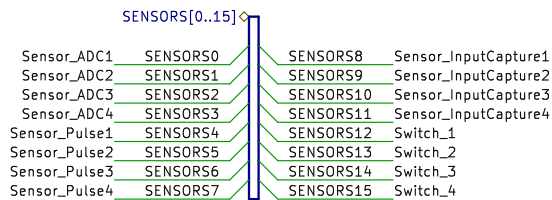
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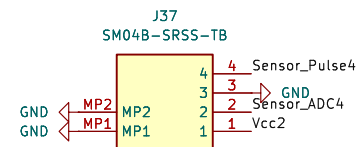
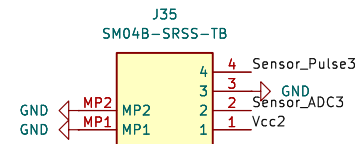
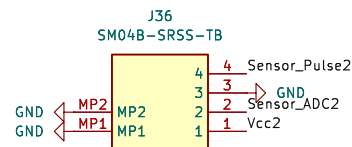
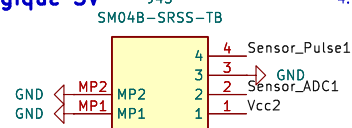
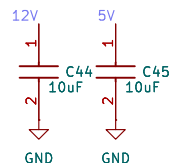


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Id: 16/17



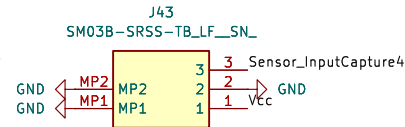
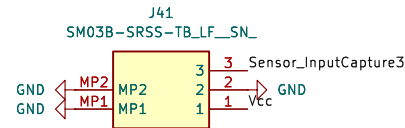
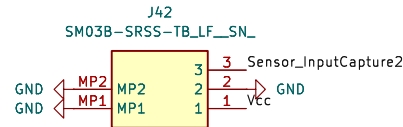
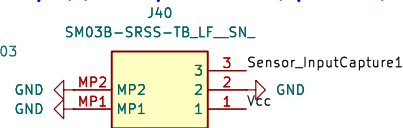
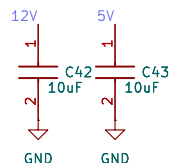
Capteurs de distance

Pour capteur B5W-LB2101-1 analogique 5V

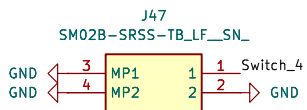
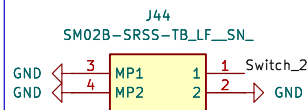
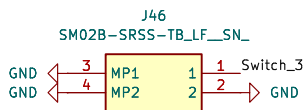
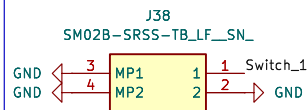


https://omronfs.omron.com/en_US/ecb/products/pdf/en-b5w_lb_series_users_manual.pdf

Pour capteur Pololu <https://www.pololu.com/product/5464>



Interrupteurs standard



TH0
GDU

Socotec Monitoring

Sheet: /Sensors/
File: Sensors.kicad_sch

Title: Robot Wall-A

Size: A4 Date: 2025-09-20

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